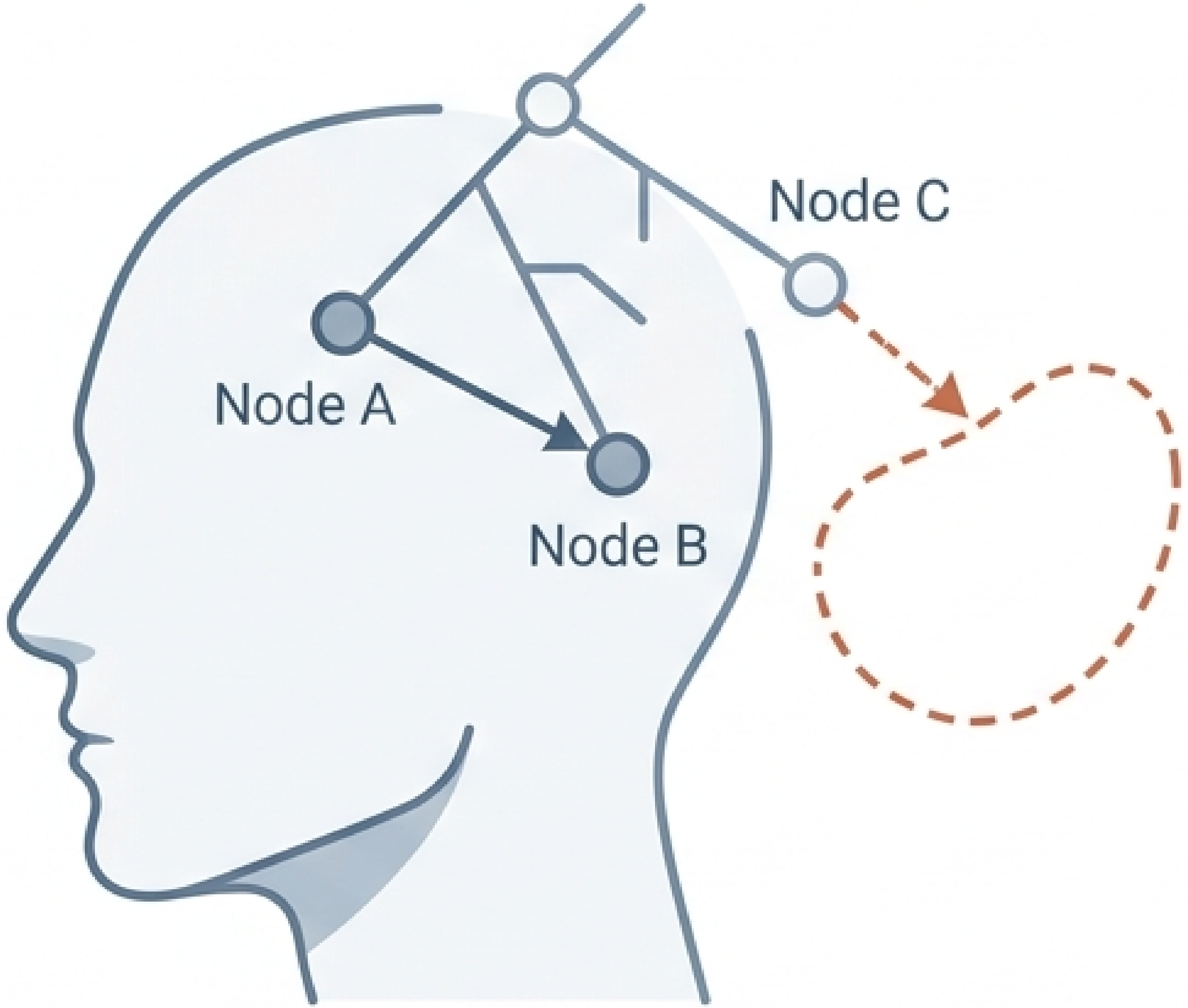
Constrained Eigenvalue Minimization

Optimal Completion of Incomplete Pairwise Comparison Matrices via Nelder-Mead

The Architecture of Human Choice

Multi-criteria decision-making frameworks rely on human pairwise comparisons. However, bounded rationality, fatigue, and lack of data mean humans rarely provide complete or perfectly consistent judgments.

The Human Element



The Mathematical Element

1	3	*	9
1/3	1	1/5	2
*	5	1	1/4
1/9	1/2	4	1

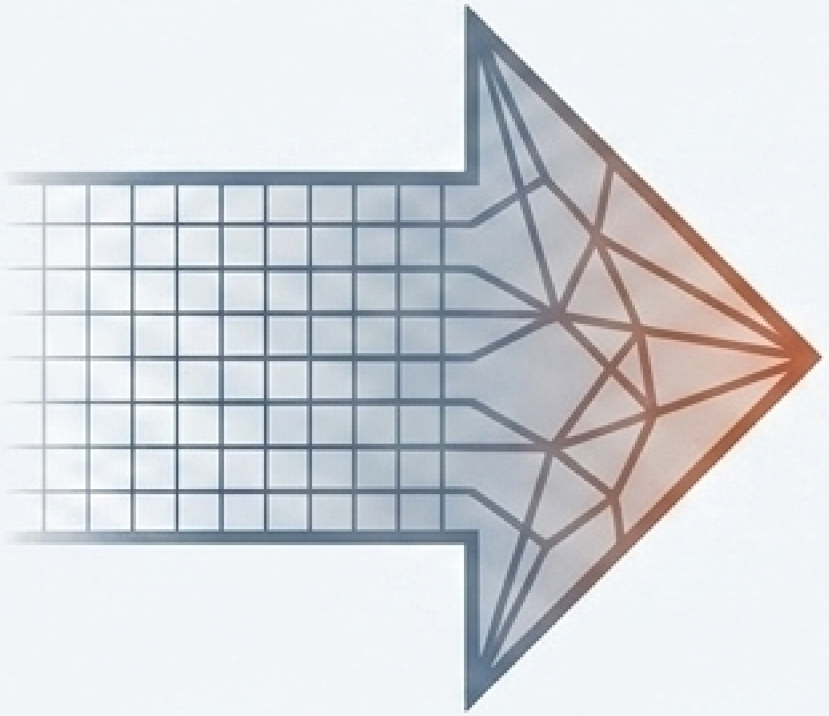
Constraint: Saaty's discrete scale restricts inputs to interval values between 1/9 and 9.

Diagnosing the Fragmented Matrix

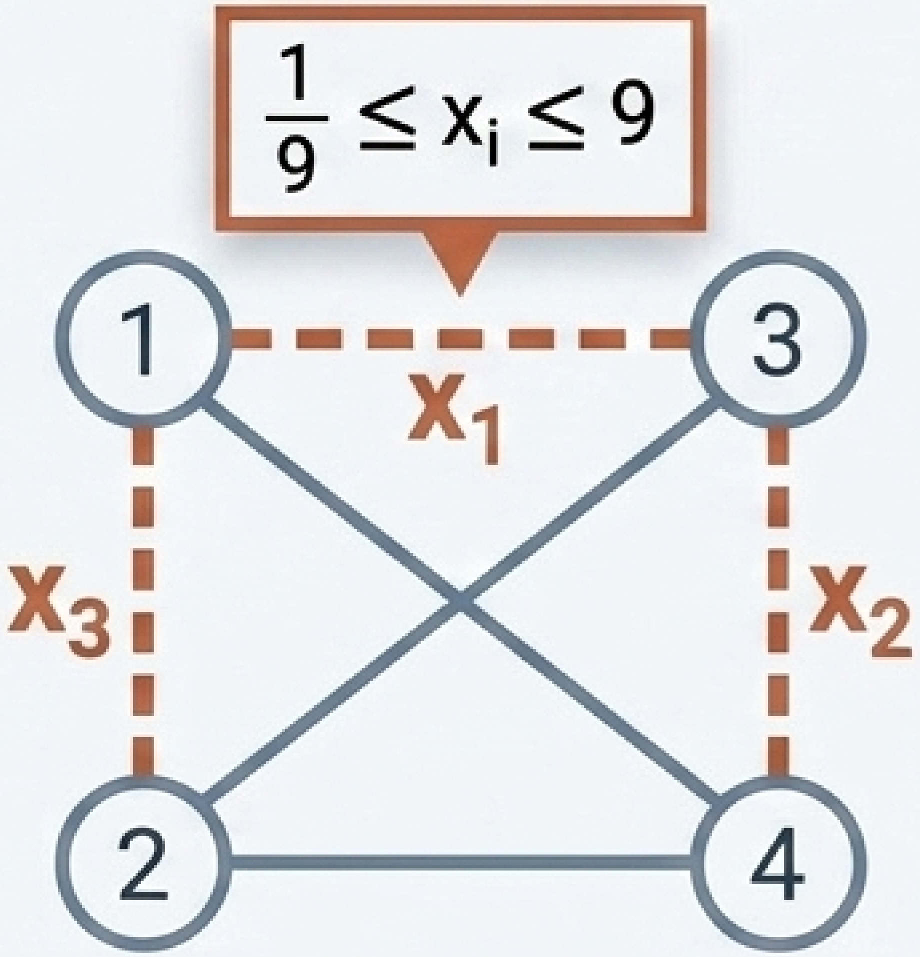
An incomplete matrix is mathematically equivalent to a fragmented network. Missing pairwise comparisons are missing edges in an undirected graph that must be calculated subject to strict intervals.

Step 1: Incomplete Matrix

1	3	x_1	9
1/3	1	x_2	2
x_3	5	1	1/4
1/9	1/2	4	1



Step 3: Fragmented Network



$G = (V, E)$

The Quest for Minimal Inconsistency

We must find the missing variables x that minimize the global inconsistency of the entire network, assuming the decision-maker is as rational as possible.

Saaty's Cut-off Rule: CR < 0.1

Saaty's Inconsistency Ratio (CR)

Objective:

$$\min \lambda_{\max}(A(\mathbf{x}))$$

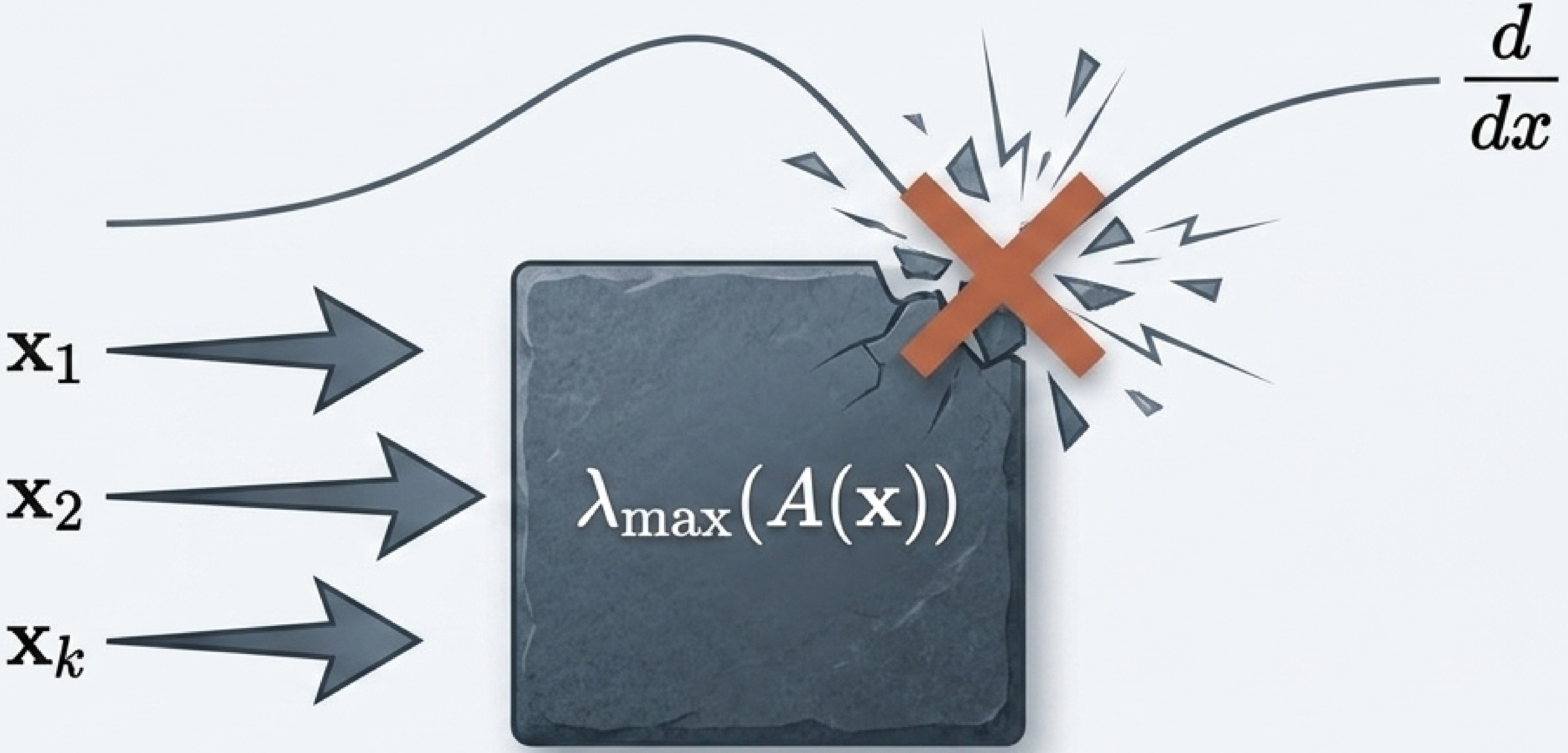
Inconsistency Formula:

$$CR = \frac{\lambda_{\max} - n}{(n-1)RI_n}$$

Where λ_{max} is the Perron eigenvalue, n is matrix size, and RI_n is the random index.

The Black Box of the Perron Eigenvalue

The maximum eigenvalue function cannot be written explicitly. Standard gradient-descent optimization fails because it requires knowing the exact slope of the terrain.

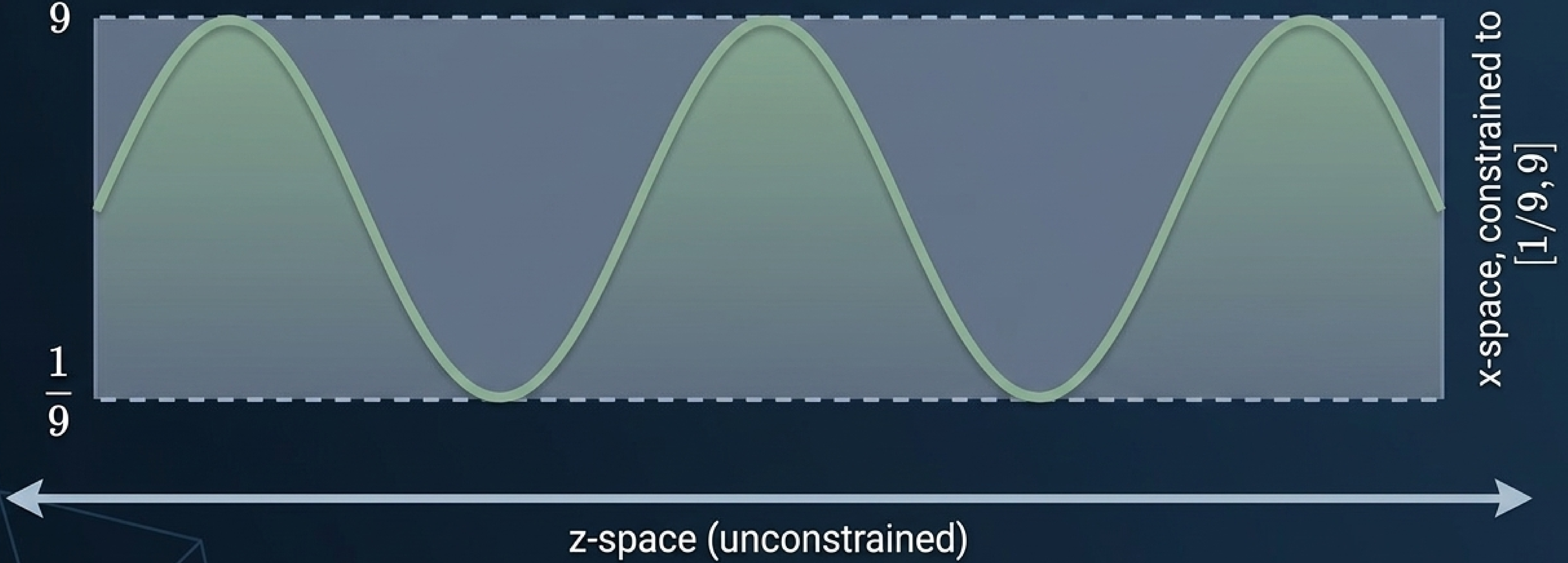


We need a method that can feel its way in the dark, navigating a non-convex topological landscape without requiring a derivative.

Innovation I: Breaking the Boundaries

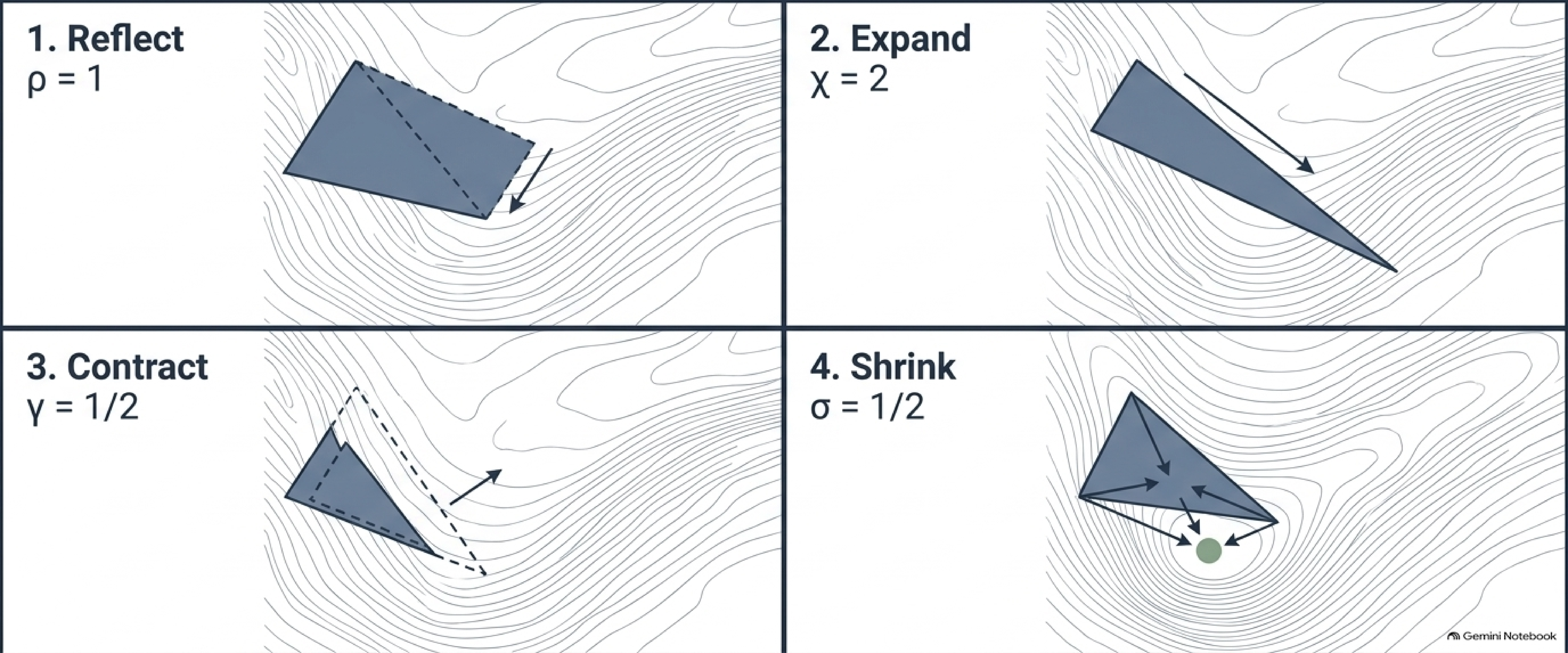
Optimization algorithms struggle with hard numerical boundaries. A trigonometric coordinate transformation elegantly converts a rigidly constrained problem into an unconstrained one.

$$x_i = l_i + \frac{1}{2}(u_i - l_i)(\sin(z_i) + 1)$$



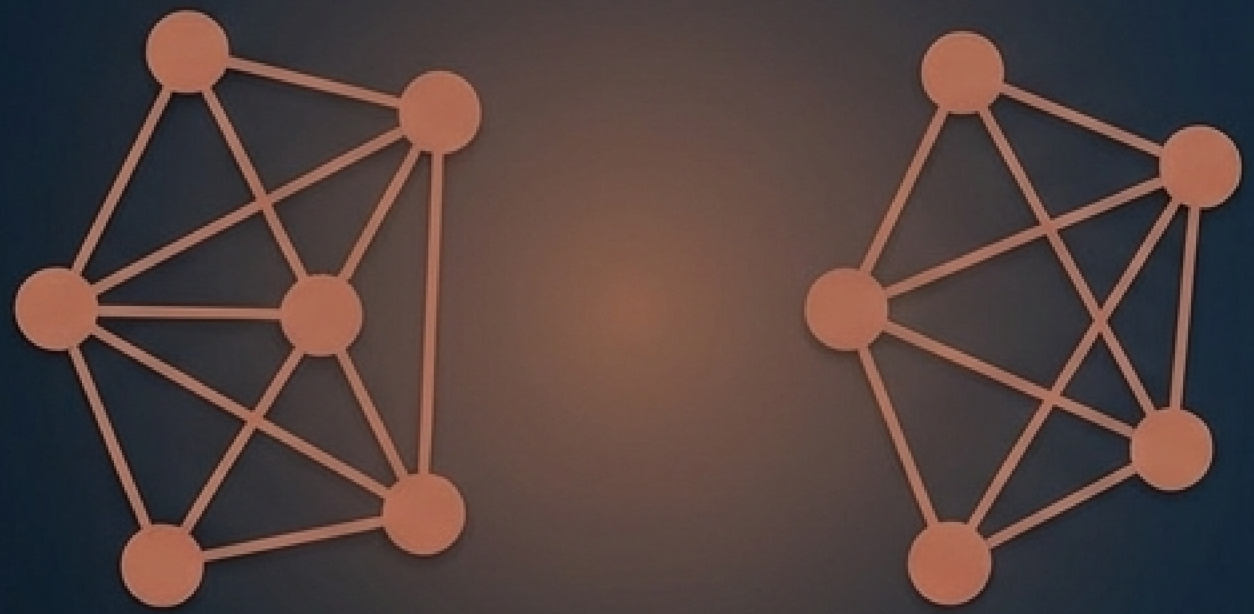
Innovation II: The Nelder-Mead Simplex

A derivative-free, direct-search method drops a geometric simplex onto the data landscape, testing points and literally crawling down the topographic hill toward optimal consistency.

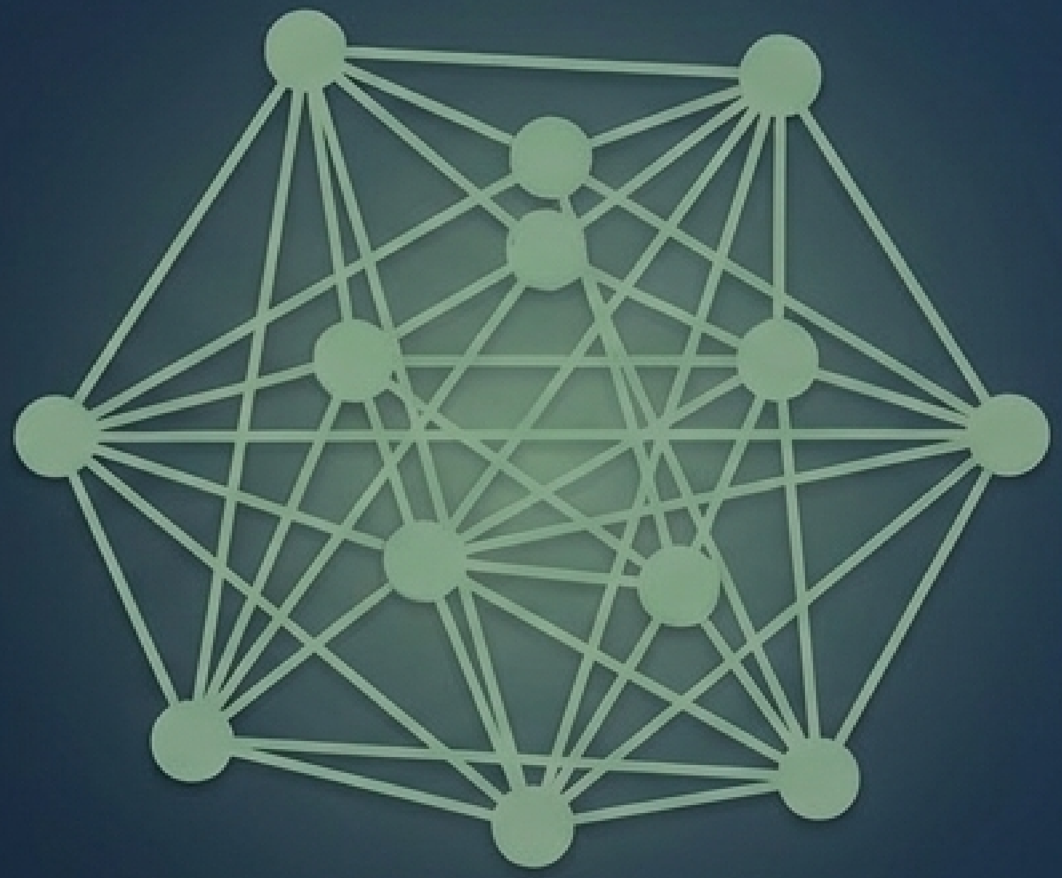


The Crucial Condition: Theorem 1

The Nelder-Mead algorithm mathematically guarantees a unique optimal completion if and only if the known data points form an entirely connected network.



Disconnected Graph
Multiple or No Solutions

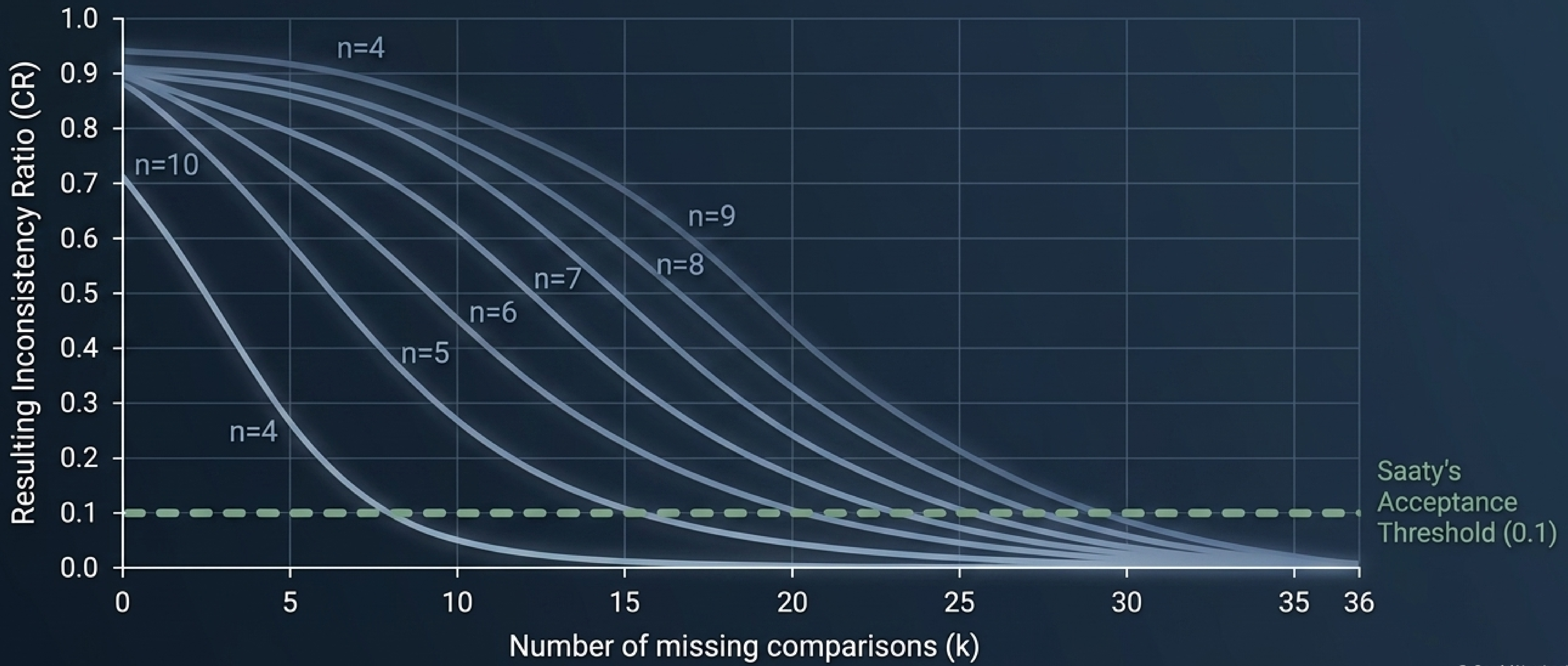


Fully Connected Graph
Unique Optimal Solution

Requirement: Missing comparisons $k \le (n-1)(n-2)/2$

Proof at Scale: 10,000 Realities

Subjected to 10,000 randomized simulations, the algorithm reliably reconstructs highly depleted matrices, consistently pulling heavy inconsistency down into acceptable ranges.



Synthesis: The Restored Matrix

Uniting trigonometric transformation with the direct-search simplex provides a highly adaptive, robust mechanism for optimally repairing human decision matrices.

Method	Handles Non-explicit Functions	Hits Exact Boundary Constraints
Logarithmic Least Squares	No	No
Cyclic Coordinate	No	Struggles
Nelder-Mead (Proposed)	Yes	Yes